

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

CS1005-1 – INTRODUCTION TO PYTHON PROGRAMMING
 (BT, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What is the purpose of the "if" statement in Python?**
 - To iterate over a list
 - To define a function
 - To perform conditional execution
 - To create a dictionary
- What is the output of the following code?**

```
x = 5
if x > 3:
    print("Hello")
else:
    print("Hi")
```

 - Hello
 - Hi
 - Neither
 - Both
- How can you add an element to the end of a list in Python?**
 - append()
 - insert()
 - extend()
 - add()
- What is the result of the expression 5 % 2?**
 - 0
 - 2
 - 1
 - 3
- Which loop is used for iterating over a sequence (e.g., a list or string) in Python?**
 - while
 - for
 - do-while
 - loop
- What is the purpose of the "elif" keyword in Python?**
 - To end the program
 - To create a new list
 - To define a function
 - To add an alternative condition to "if"
- Which data structure in Python is mutable?**
 - Tuple
 - List
 - String
 - Set
- How do you declare a dictionary in Python?**
 - { }
 - []
 - ()
 - < >
- Which operator is used for exponentiation in Python?**
 - **
 - ^
 - //
 - *
- What does the "return" statement do in a Python function?**
 - Prints a value to the console
 - Exits the function immediately
 - Returns a value to the caller
 - Raises an exception
- How do you remove an item from a list by its value in Python?**
 - remove()
 - delete()
 - pop()
 - discard()

12. What is the output of the code snippet below?
`numbers = [1, 2, 3, 4, 5]`
`squared = [x**2 for x in numbers]`
`print(squared)`
 A) [1, 4, 9, 16, 25] B) [1, 2, 3, 4, 5]
 C) [1, 8, 27, 64, 125] D) [2, 4, 6, 8, 10]
13. How do you access the value associated with a specific key in a dictionary?
 A) Using the index B) Using a loop
 C) Using the key itself D) Using the value() function
14. Which of the following is a valid way to comment out multiple lines of code in Python?
 A) // Comment B) /* Comment */
 C) # Comment D) "" Comment ""
15. What does the "continue" statement do in a loop?
 A) Terminates the loop B) Skips the current iteration and proceeds to the next
 C) Restarts the loop from the beginning D) Does nothing
16. Which of the following is not a valid data type in Python?
 A) int B) float
 C) char D) bool
17. How do you check the length of a list in Python?
 A) size() B) length()
 C) count() D) len()
18. What is the output of the following code snippet?
`def add(x, y):`
 `return x + y`

`result = add(2, 3)`
`print(result)`
 A) "5" B) 5
 C) 23 D) "23"
19. What does the range() function return in Python ?
 A) A list of integers B) A tuple of integers
 C) A range object D) A string
20. From the given syntax, which of the following is the correct syntax to create a dictionary?
 A) Dict=[1:"hi",2:"hello" B) Dict={ 1:"hi",2:"hello"}
 C) Dict={ 1:"hi",2:"hello"} D) Dict=[1:"hi",2:"hello"]

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare and contrast primary and secondary memory in computer.	6	L*2	1	1
	b) Define the following terms with example.				
	i. Operator				
	ii. Datatypes	4	L2	2	1
	c) Develop a python program to find GCD and LCM of given numbers.	6	L3	2	2
2.	a) Design an algorithm and flowchart to find the factorial of given number.	7	L3	1	2
	b) Illustrate loops in python with an example.	9	L2	1	1
3.	a) Explain steps for program development with a neat diagram.	8	L2	1	1
	b) Explain elif conditional statement in python. Develop a python program to find the roots of a quadratic equation $ax^2 + bx + c = 0$	8	L3	2	2

Unit – II

4.	a)	Define list and explain accessing of list elements with code snippet.	5	L2	3	1
	b)	Design a python program to perform binary search for unsorted elements by designing two separate functions to perform search and sort. Hint: sort() built-in function can be used.	8	L3	4	2
	c)	List and explain any three advantages of functions in python.	3	L2	4	1
5.	a)	Compare and contrast difference between list and tuples.	5	L2	3	1
	b)	Define dictionaries. Illustrate any three built-in function on dictionaries with example.	6	L2	3	1
	c)	Design python program to extract the keys from a given dictionary using function approach.	5	L3	4	2
6.	a)	Write a Python program to read a list of n integers (positive as well as negative). Create 2 new lists, one having all positive numbers and others having all negative numbers from the given list. Print all 3 lists.	10	L3	3	2
	b)	Explain any three list operations in python with example.	6	L2	3	1

Unit – III

7.	a)	Write a Python program that takes a string input from the user and performs the following operations:				
	i)	Count the number of characters in the string and display the result.				
	ii)	Convert all characters in the string to uppercase and display the result.				
	iii)	Check if the string contains the word (case-insensitive) and display the result as True or False.				
	iv)	Split the string into a list of words and display the result.				
	v)	Join the list of words using a hyphen as a separator and display the result.				
	vi)	Reverse the order of the characters in the string and display the result.	10	L3	5	2
	b)	Illustrate the slicing of string with an example.	6	L2	5	1
8.	a)	Define file. Illustrate read and write operation in file with a code snippet.	8	L2	5	1
	b)	Write a function display_words() in python to read lines from a text file "story.txt", and display those words, which are less than 4 characters.	8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
